

Tracking attractors via discrete rough paths

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Numerical attractors

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Gradient decent method

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta), \quad Q \in C^1. \quad (1.1)$$

- Global solution: $\theta^* \in \arg \min_{\theta \in \mathbb{R}^d} Q(\theta) \Rightarrow \nabla Q(\theta^*) = 0$. This leads to an approximation scheme to achieve θ^* , which is the zero solution of $\nabla Q(\theta) = 0$

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t), \quad (1.2)$$

for the learning rate $\eta > 0$. (1.2) is called Gradient descent method.

- When η small enough, (1.2) can also be viewed as the Euler scheme for solving the ODE

$$\dot{\theta} = -\nabla Q(\theta), \quad (1.3)$$

where one attempts to solve for the steady state θ^* of (1.3).

Stochastic gradient descent

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta) = \min_{\theta \in \mathbb{R}^d} \frac{1}{m} \sum_{i=1}^m Q^{(i)}(\theta), \quad Q^{(i)} \in C^1. \quad (1.4)$$

When m large enough, (1.2) is replaced by: at each step t choose a random set Ω_t of $\{1, \dots, m\}$ (with or without replacement) and then updating

$$\theta_{t+1} := \theta_t - \eta \frac{1}{|\Omega_t|} \sum_{i \in \Omega_t} \nabla Q^{(i)}(\theta_t) = \theta_t - \eta \nabla \tilde{Q}_t(\theta_t) \quad (1.5)$$

No matter how random the choice, can rewrite SGD recursion

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t) + \eta^\alpha \epsilon_t(\theta_t). \quad (1.6)$$

- (1.6) can also be viewed as the Euler scheme of the "limiting equation"

$$d\theta = -\nabla Q(\theta)dt + \sigma(\theta)dX_t, \quad (1.7)$$

where $\sigma\sigma^T = \eta^{2\alpha}\mathbb{E}[\epsilon\epsilon^T|\theta_t = \theta]$.

- If Ω_t i.i.d. then CLT gives $\alpha = \frac{1}{2}$ and X -Gaussian. But not necessary i.i.d.
- Questions: How does the distribution of (1.6) converge to some limit? And which calculus is used to interpret and solve the limiting equation?
- Work by Hairer et al. on ergodicity if X is fractional Brownian motion. But we prefer an approach which is purely discrete.

Residual Neural Networks via discrete rough paths

- In a plain neural network, the input vector y_{i+1} of the $(i+1)$ -th hidden layer is given by an application of the weights and the activation function to the input of the previous hidden layer.

$$y_{i+1} = \sigma(\theta_i y_i),$$

where $\sigma : \mathbb{R}^{d_{i+1}} \rightarrow \mathbb{R}^{d_{i+1}}$ and $\theta_i \in \mathbb{R}^{d_{i+1} \times d_i}$. In the ResNet approach, this is modified so that the output to the next hidden layer is given as the sum of the input to the previous layer, plus the previous operations

$$y_{i+1} = y_i + \sigma(\theta_i y_i). \quad (1.8)$$

- View (1.8) as a step of the Euler scheme for a controlled ODE

$$\dot{y}_t = \sigma(\theta_t y_t), \quad y_0 \in \mathbb{R}^{d_0}. \quad (1.9)$$

- ResNet with finitely many layers:

$$y_{k+1} = y_k + \sum_{\mu=1}^d f^{(\mu)}(y_k)(x_{k+1}^{(\mu)} - x_k^{(\mu)}). \quad (1.10)$$

- Stability of ResNet (Bayer-Friz-Tapia, 2023): $2 \leq p < 3$ and $f = (f^\mu) \in C_b^3$

$$\begin{aligned} \sup_{k=0..n} \|y_k - \tilde{y}_k\| &\leq C'_{p,n} \exp \left\{ C_{p,n} \|f\|_{C_b^3}^p \left(\|\mathbf{x}\|_{p,[0,n]}^p + \|\tilde{\mathbf{x}}\|_{p,[0,n]}^p \right) \right\} \\ &\quad \times (\|\xi - \tilde{\xi}\| + \rho_p(\mathbf{x}, \tilde{\mathbf{x}})) \end{aligned} \quad (1.11)$$

Discrete functions and norms

Let $[a, b] \subset \mathbb{R}$ and $\Pi = \{a = t_0 < t_1 < \dots < t_n = b\}$ be an arbitrary (finite) set of points of $[a, b]$. We introduce various norms which are natural discrete versions of the continuous ones

$$\|y\|_{\infty, \Pi} := \sup_{t_i \in \Pi} \|y_{t_i}\|;$$

$$\|y\|_{p, \Pi} := \sup_{t_i^* \in \Pi, 0 \leq i \leq r, t_0^* < t_1^* \dots < t_r^*} \left(\sum_{i=0}^{r-1} \|y_{t_i^*} - y_{t_{i+1}^*}\|^p \right)^{1/p};$$

$$\|y\|_{p, \Pi} = \|y_a\| + \|y\|_{p, \Pi};$$

$$\|F\|_{p, \Pi} := \sup_{t_i^* \in \Pi, 0 \leq i \leq r, t_0^* < t_1^* \dots < t_r^*} \left(\sum_{i=0}^{r-1} |F_{t_i^*, t_{i+1}^*}|^p \right)^{1/p},$$

$$\|F\|_{\infty, \Pi} := \sup_{a \leq s \leq t \leq b} \|F_{s, t}\|.$$

For a path ξ we write $\xi_{s, t} = \xi_t - \xi_s$.

Discrete controls

A non negative function ω defined on $\Delta\Pi := \{(s, t) \in \Pi^2 | s \leq t\}$ is called a *discrete control function* on Π if it vanishes on the diagonal, i.e. $\omega_{s,s} = 0, \forall s \in \Pi$, and is superadditive, i.e. for all $s \leq u \leq t$ in Π

$$\omega_{s,u} + \omega_{u,t} \leq \omega_{s,t}.$$

As an example, if F vanishes on the diagonal of Π , $\omega_{s,t} := \|F\|_{q,\Pi[s,t]}^q, (s, t) \in \Delta\Pi$ is a discrete control. If ω is a discrete control on $[a, b]$ and $|y_{t_k} - y_{t_l}| \leq \omega_{t_k, t_l}^{1/p}$ for all $t_k, t_l \in \Pi, p \geq 1$ then

$$\|y\|_{p,\Pi[a,b]} \leq \omega_{a,b}^{1/p}.$$

Discrete rough paths

- For given $[a, b]$ and Π a (finite) discrete time set in $[a, b]$ as above, let $x : \Pi \rightarrow \mathbb{R}^m$ and $\mathbb{X}(\cdot, \cdot) : \Pi^2 \rightarrow \mathbb{R}^{m \times m}$ be discrete functions defined on Π and Π^2 respectively. One then says that x can be lifted to a discrete rough path $\mathbf{x} = (x, \mathbb{X})$ if it satisfies Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad s \leq u \leq t; \quad s, u, t \in \Pi. \quad (2.1)$$

One can then furnish $\mathbf{x}$ with a semi-norm

$$\|\mathbf{x}\|_{p, \Pi[a,b]} = \left(\|x\|_{p, \Pi[a,b]}^p + \|\mathbb{X}\|_{p/2, \Pi[a,b]}^{p/2} \right)^{1/p}. \quad (2.2)$$

- Example:

$$\mathbb{X}_{t_k, t_l}^{\mu, \nu} := \sum_{j=k}^{l-1} x_{t_k, t_j}^{(\mu)} x_{t_j, t_{j+1}}^{(\nu)}.$$

Discrete sewing lemma

Let $\Pi = \{t_i : 0 \leq i \leq n, a = t_0 < t_1 < \dots < t_n = b\}$ be a finite partition of $[a, b]$ and F be an function defined on $\Delta\Pi$, vanished on the diagonal, i.e. $F_{s,s} = 0, \forall s \in \Pi$. Put

$$\begin{aligned}(\delta F)_{s,u,t} &:= F_{s,t} - F_{s,u} - F_{u,t}, \quad s \leq u \leq t \in \Pi, \\ I_{k,l} &:= \sum_{k \leq j \leq l-1} F_{t_j, t_{j+1}} - F_{t_k, t_l}, \quad t_k \leq t_l \in \Pi.\end{aligned}$$

Assume that for a discrete control ω on Π and a number $\lambda > 1$ we have for all $s \leq u \leq t$ in Π the following inequality

$$\|(\delta F)_{s,u,t}\| \leq \omega_{s,t}^\lambda. \quad (2.3)$$

Then there exists a constant $K > 0$ depending only on λ such that

$$\|I_{k,l}\| \leq K \omega_{t_k, t_l}^\lambda, \quad \forall t_k \leq t_l \in \Pi. \quad (2.4)$$

Proof of discrete sewing lemma

Fix a $\theta \geq \frac{2}{1-2^{1-\lambda}} > 2$. We are going to prove (2.4) by induction for $l - k$. It is easy to check that (2.4) holds trivially for $l - k = 0, 1$. Assume it still holds for all k, l such that $k \leq l < m$. We will prove that (2.4) is true for $l = m$. Define

$$l := \max\{k \leq i \leq m : \omega_{t_k, t_i} \leq \frac{1}{2}\omega_{t_k, t_m}\}.$$

By definition, $\omega_{t_k, t_l} \leq \frac{1}{2}\omega_{t_k, t_m}$ and $\omega_{t_k, t_{l+1}} > \frac{1}{2}\omega_{t_k, t_m}$. It follows that $\omega_{t_{l+1}, t_m} < \frac{1}{2}\omega_{t_k, t_m}$ due to super additivity of ω . Since $l_{l, l+1} = 0$, it follows that

$$\begin{aligned} l_{k, m} &= l_{k, l} + l_{l, m} - (\delta F)_{t_k, t_l, t_m} \\ &= l_{k, l} + l_{l, l+1} + l_{l+1, m} - (\delta F)_{t_l, t_{l+1}, t_m} - (\delta F)_{t_k, t_l, t_m} \\ &= l_{k, l} + l_{l+1, m} - (\delta F)_{t_l, t_{l+1}, t_m} - (\delta F)_{t_k, t_l, t_m}. \end{aligned}$$

By inductive hypothesis and assumption of δF we obtain

$$\begin{aligned}
\|I_{k,m}\| &\leq \|I_{k,l}\| + \|I_{l+1,m}\| + \|(\delta F)_{t_l, t_{l+1}, t_m}\| + \|(\delta F)_{t_k, t_l, t_m}\| \\
&\leq \theta \omega_{t_k, t_l}^\lambda + \theta \omega_{t_{l+1}, t_m}^\lambda + \omega_{t_l, t_m}^\lambda + \omega_{t_k, t_m}^\lambda \\
&\leq 2^{1-\lambda} \theta \omega_{t_k, t_m}^\lambda + 2 \omega_{t_k, t_m}^\lambda \\
&\leq (2^{1-\lambda} \theta + 2) \omega_{t_k, t_m}^\lambda \\
&\leq \theta \omega_{t_k, t_m}^\lambda.
\end{aligned}$$

This completes the proof.

Corollary: Let $\Pi = \{t_i : 0 \leq i \leq n, a = t_0 < t_1 < \dots < t_n = b\}$ be a finite partition of $[a, b]$. Consider a discrete system defined on Π

$$y_{t_{j+1}} = y_{t_j} + F_{t_j, t_{j+1}} + \epsilon_{t_j, t_{j+1}}, \quad y_{t_0} = y^* \in \mathbb{R}^d, \quad j = 0, 1, \dots, n-1, \quad (2.5)$$

where $F : \Pi^2 \rightarrow \mathbb{R}^d$, $\epsilon : \Pi^2 \rightarrow \mathbb{R}^d$. Assume that

- (i) There exists a discrete control ω such that (2.3) is satisfied for F ;
- (ii) There exists a discrete control function $\omega^{(0)}$ such that

$$|\epsilon_{t_j, t_{j+1}}| \leq \omega_{t_j, t_{j+1}}^{(0)}.$$

Then there exists a constant $K > 0$ such that for any $r \geq 1$

$$\|y\|_{r, \Pi[a, b]} \leq K \omega_{a, b}^\lambda + \|F\|_{r, \Pi[a, b]} + \omega_{a, b}^{(0)}. \quad (2.6)$$

Greedy sequence of times

Continuous time version: see Cass-Litterer-Lyons, 2013.

Given a fixed $\gamma > 0$, assign the starting time $\tau_0 = a$. For each $m \in \mathbb{N}$, assume $\tau_m = t_k$ is determined. Then one can define τ_{m+1} by the following rule:

- if $\sum_{\omega \in \mathcal{S}} \omega_{t_k, t_{k+1}}^{\beta_\omega} > \gamma$ then set $\tau_{m+1} := t_{k+1}$;
- else set $\tau_{m+1} := \sup\{t_l \in (t_k, b] : \sum_{\omega \in \mathcal{S}} \omega_{t_k, t_l}^{\beta_\omega} \leq \gamma\}$.

Define $N(\gamma, [a, b])$ to be the number of times τ_m in $[a, b]$. From the definition,

$$\sum_{\omega \in \mathcal{S}} \omega_{\tau_m, \tau_{m+2}}^{\beta_\omega} > \gamma.$$

By taking both sides to the power of $\frac{1}{\beta}$ where

$\beta := \min\{\beta_\omega : \omega \in \mathcal{S}\} \leq 1$, and using the Jensen's inequality, one obtains

$$\gamma^{\frac{1}{\beta}} < \left(\sum_{\omega \in \mathcal{S}} \omega_{\tau_m, \tau_{m+2}}^{\beta_\omega} \right)^{\frac{1}{\beta}} \leq (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \left(\sum_{\omega \in \mathcal{S}} \omega_{\tau_m, \tau_{m+2}}^{\frac{\beta_\omega}{\beta}} \right). \quad (2.7)$$

As a result, taking the sum of both sides of (2.7) as m going from 0 to $N(\gamma, [a, b]) - 3$ (whenever $N(\gamma, [a, b]) \geq 3$) and using the fact that all elements in the square brackets of (2.7) are controls, one obtains

$$\begin{aligned} [N(\gamma, [a, b]) - 2]\gamma^{\frac{1}{\beta}} &< \sum_{m=0}^{N(\gamma, [a, b]) - 3} (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \left(\sum_{\omega \in \mathcal{S}} \omega^{\frac{\beta\omega}{\beta}} \right) \\ &< (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \sum_{\omega \in \mathcal{S}} 2\omega^{\frac{\beta\omega}{\beta}}_{a,b}. \end{aligned}$$

Hence, $N(\gamma, [a, b])$ can be estimated as

$$N(\gamma, [a, b]) < 2 + \frac{2}{\gamma^{\frac{1}{\beta}}} (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \sum_{\omega \in \mathcal{S}} \omega^{\frac{\beta\omega}{\beta}}_{a,b}. \quad (2.8)$$

- For Young equations, one considers $\omega_{s,t}^{(1)} = C_f(t-s)$ with $\beta_1 = 1$ and $\omega_{s,t}^{(2)} = L_g^p \mathbb{W}_{p,\Pi[s,t]}^p$ with $\beta_2 = \frac{1}{p}$ and $\mathcal{S} = \{\omega^{(1)}, \omega^{(2)}\}$. Then $\beta = \frac{1}{p}$ and (2.8) has the form

$$N(\gamma, [a, b]) < 2 + \frac{2}{\gamma^p} 3^{p-1} \left[C_f^p (b-a)^p + L_g^p \mathbb{W}_{p,\Pi[a,b]}^p \right]. \quad (2.9)$$

- For rough equations, one consider $\mathcal{S} = \{\omega^{(1)}, \omega^{(2)}, \omega^{(3)}\}$, $\omega_{s,t}^{(1)} = t-s$, $\beta_1 = 1$, $\omega_{s,t}^{(2)} = L_g^p \mathbb{W}_{p,\Pi[s,t]}^p$, $\beta_2 = \frac{1}{p}$ and $\omega_{s,t}^{(3)} = L_g^q \mathbb{W}_{q,\Pi[s,t]}^q$, $\beta_3 = \frac{1}{q} = \frac{2}{p}$. Then $\beta = \frac{1}{p}$ and (2.8) has the form

$$N(\gamma, [a, b]) < 2 + \frac{2}{\gamma^p} 4^{p-1} \left[C_f^p (b-a)^p + L_g^p \mathbb{W}_{p,\Pi[a,b]}^p + L_g^q \mathbb{W}_{q,\Pi[a,b]}^q \right]. \quad (2.10)$$

Rough difference equations

Discrete time set $\Pi = \{t_k\}_{k \in \mathbb{N}}$ defined by

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k, t_{k+1}}(y, \mathbf{x}), \quad (3.1)$$

$$F_{s,t}(y, \mathbf{x}) = \begin{cases} g(y_s) \mathbb{X}_{s,t}, & \mathbf{x} = x \\ g(y_s) \mathbb{X}_{s,t} + Dg(y_s)g(y_s) \mathbb{X}_{s,t}, & \mathbf{x} = (x, \mathbb{X}) \end{cases} \quad (3.2)$$

(H_f) $f \in C(\mathbb{R}^d, \mathbb{R}^d)$ of linear growth, i.e.

$$\exists C_f > 0 : \quad \|f(y)\| \leq C_f \|y\| + \|f(0)\|, \quad \forall y \in \mathbb{R}^d; \quad (3.3)$$

(H_g) either $g \in C_b^2(\mathbb{R}^d, \mathbb{R}^{d \times m})$ such that

$$\|g\|_\infty, C_g := \max \left\{ \|Dg\|_\infty, \|D^2g\|_\infty \right\} < \infty, \quad (3.4)$$

or $g(y) = Cy + g(0)$, $C \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^{d \times m})$ s.t. $\|C\| \leq C_g$. For YDE, $g \in C^1$ with bounded derivative.

- If $F_{s,t} = g(y_s)x_{s,t}$ then by assigning $L_g := C_g$ one obtains

$$\|\delta F_{s,u,t}\| \leq C_g \|y_{s,u}\| \|x_{u,t}\| \leq L_g \|y\|_{p,\Pi[s,t]} \|x\|_{p,\Pi[s,t]}. \quad (3.5)$$

- If $F_{s,t} = g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}$ where $g \in C_b^2$ or $g(y) = Cy + g(0)$ then a direct computation shows that

$$\begin{aligned} (\delta F)_{s,u,t} &= \left[- \int_0^1 Dg(y_s + \eta y_{s,u})(g(y_s)x_{s,u} + R_{s,u}^y) d\eta \right] x_{u,t} \\ &\quad + Dg(y_s)g(y_s)x_{s,u} \otimes x_{u,t} \\ &\quad + [Dg(y_s)g(y_s) - Dg(y_u)g(y_u)]\mathbb{X}_{u,t}. \end{aligned}$$

From the assumption $(\mathbf{H}_g)$, $Dg(y)g(y)$ is globally Lipschitz continuous with constant L_g^2 given by

$$L_g = \begin{cases} \sqrt{C_g^2 + C_g \|g\|_\infty} & \text{if } g \in C_b^2; \\ C_g & \text{if } g(y) = Cy + g(0). \end{cases}$$

- Define

$$R_{s,t}^y := y_{s,t} - g(y_s)x_{s,t}, \quad s, t \in \Pi.$$

As a result, it is easy to check that for a generic constant $K > 0$

$$\begin{aligned}
 \|(\delta F)_{s,u,t}\| &\leq L_g \|R_{s,u}^y\| \|x_{u,t}\| + L_g^2 \|y_{s,u}\| (\|x_{s,u} \otimes x_{u,t}\| + \|\mathbb{X}_{u,t}\|) \\
 &\leq L_g \|R_{s,u}^y\| \|x_{u,t}\| \\
 &\quad + L_g^2 \|y_{s,u}\| (\|\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t}\| + \|\mathbb{X}_{u,t}\|) \\
 &\leq K (L_g \|R^y\|_{q,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} \\
 &\quad + L_g^2 \|y\|_{p,\Pi[s,t]} \|\mathbb{X}\|_{q,\Pi[s,t]}). \tag{3.6}
 \end{aligned}$$

- It is easy to check that $\left(\|y\|_{p,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} \right)^{\frac{p}{2}}$ for $1 \leq p < 2$, and $(\|R^y\|_{q,\Pi[s,t]} \|x\|_{p,\Pi[s,t]})^{\frac{p}{3}}$ and $(\|y\|_{p,\Pi[s,t]} \|\mathbb{X}\|_{q,\Pi[s,t]})^{\frac{p}{3}}$ for $2 \leq p < 3$, are control functions. Hence from now on, we can write symbolically $\omega_{s,t}^{\lambda_F}(\delta F)$ to indicate the right hand side of (3.5) or (3.6), where $\lambda_F = \frac{2}{p}$ for $1 < p < 2$ and $\lambda_F = \frac{3}{p}$ for $2 \leq p < 3$.

Lemma

Lemma: Let $\{\tau_i\}_{i=\overline{0,m}} \subset \Pi$ be an sub partition of $[a, b]$. The following estimates hold

$$\begin{aligned} \|y\|_{p,\Pi} &\leq m^{\frac{p-1}{p}} \sum_{k=0}^{m-1} \|y\|_{p,\Pi[\tau_k, \tau_{k+1}]}; \\ \|R^y\|_{q,\Pi} &\leq (2m)^{\frac{q-1}{q}} \left(\sum_{k=0}^{m-1} \|R^y\|_{q,\Pi[\tau_k, \tau_{k+1}]} + C_g m^{\frac{1}{p}} \|x\|_{p,\Pi} \sum_{k=0}^{m-1} \|y\|_{p,\Pi[\tau_k, \tau_{k+1}]} \right). \end{aligned} \quad (3.7)$$

Proof of Lemma: Observe that

$$\begin{aligned} \|(\delta R^y)_{s,u,t}\| &\leq C_g \|y_{s,u}\| \|x_{u,t}\| \\ \Rightarrow \|R^y_{s,t}\| &\leq \|R^y_{s,u}\| + \|R^y_{u,t}\| + C_g \|y_{s,u}\| \|x_{u,t}\|, \quad \forall s \leq u \leq t. \end{aligned} \quad (3.8)$$

Proof of Lemma

Now given any finite partition $\Pi^* \subset \Pi$ of $[a, b]$, there exists a subsequence $\tau_l \leq \dots \leq \tau_L$ of $\{\tau_i\}_{i=0}^m$ in the interval $[s, t]$ for every consecutive points s, t of Π^* . Then one can apply (3.8) to obtain

$$\|R_{s,t}^y\| \leq \|R_{s,\tau_l}^y\| + \|R_{\tau_l,\tau_{l+1}}^y\| + \dots + \|R_{\tau_L,t}^y\| + C_g \|x\|_{p,\Pi[s,t]} \left(\|y_{s,\tau_l}\| + \dots + \|y_{\tau_L,t}\| \right).$$

As a result, one can use the fact that $\|R^y\|_{q,\Pi[s,t]}^q$ and $\|y\|_{p,\Pi[s,t]}^p$ are controls and apply Jensen's inequality to obtain

$$\begin{aligned} & \left(\sum_{s,t \in \Pi^*} \|R_{s,t}^y\|^q \right)^{\frac{1}{q}} \\ & \leq (2m)^{\frac{q-1}{q}} \left\{ \left(\sum_{s,t \in \Pi^*} (\|R_{s,\tau_l}^y\|^q + \|R_{\tau_l,\tau_{l+1}}^y\|^q + \dots + \|R_{\tau_L,t}^y\|^q) \right)^{\frac{1}{q}} \right. \\ & \quad \left. + C_g \left(\sum_{s,t \in \Pi^*} \|x\|_{p,\Pi[s,t]}^q (\|y_{s,\tau_l}\|^q + \dots + \|y_{\tau_L,t}\|^q) \right)^{\frac{1}{q}} \right\} \\ & \leq (2m)^{\frac{q-1}{q}} \left\{ \sum_{k=0}^{m-1} \|R^y\|_{q,\Pi[\tau_k,\tau_{k+1}]} + C_g m^{\frac{1}{p}} \|x\|_{p,\Pi} \sum_{k=0}^{m-1} \|y\|_{p,\Pi[\tau_k,\tau_{k+1}]} \right\}. \end{aligned}$$

Solution estimate

Theorem: There exist polynomials $\xi_i(L_g \|\mathbf{x}\|_{p, \Pi[a, b]}), i = 1, 2, 3$ of $L_g \|\mathbf{x}\|_{p, \Pi[a, b]}$ such that the solution of the rough difference equation (3.1) satisfies

$$\begin{aligned} (i), \quad & \|y\|_{\infty, \Pi[a, b]} \leq \|y_a\| e^{\xi_1(L_g \|\mathbf{x}\|_{p, \Pi[a, b]})} + C_0 e^{\xi_2(L_g \|\mathbf{x}\|_{p, \Pi[a, b]})} - C_0; \\ (ii), \quad & \|(y, R^y)\|_{p, \Pi[a, b]} \leq (\|y_a\| + C_0) e^{\xi_3(L_g \|\mathbf{x}\|_{p, \Pi[a, b]})}. \end{aligned} \quad (3.9)$$

In case g is bounded, ξ_1 can be chosen independently of $\mathbf{x}$.

Proof of (3.9)

Starting from (3.1) for two consecutive moments t_k, t_{k+1} it follows from the discrete sewing lemma that

$$\begin{aligned}\|y_{s,t} - F_{s,t}\| &\leq \left\| \sum_{t_k, t_{k+1} \in \Pi[s,t]} y_{t_k, t_{k+1}} - F_{s,t} \right\| \\ &\leq \sum_{t_k, t_{k+1} \in \Pi[s,t]} \|f(y_{t_k})\| (t_{k+1} - t_k) \\ &\quad + \left\| \sum_{t_k, t_{k+1} \in \Pi[s,t]} F_{t_k, t_{k+1}} - F_{s,t} \right\| \\ &\leq \|f(y)\|_{\infty, \Pi[s,t]} (t - s) + K \omega_{s,t}^{\lambda_F}(\delta F).\end{aligned}$$

As a result,

$$\|y - F\|_{p, \Pi[s,t]} \leq \|f(y)\|_{\infty, \Pi[s,t]} (t - s) + K \omega_{s,t}^{\lambda_F}(\delta F). \quad (3.10)$$

Young DE

First, we consider the Young difference equation, i.e. $F_{s,t} = g(y_s)x_{s,t}$ and $L_g = C_g$. It then follows from (3.5) and (3.10) that there exists generic constants K and C_0 such that

$$\begin{aligned}
 \|y\|_{p,\Pi[s,t]} &\leq \|f(y)\|_{\infty,\Pi[s,t]}(t-s) + \|g(y)\|_{\infty,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} \\
 &\quad + KL_g \|y\|_{p,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} \\
 &\leq (C_f \|y\|_{\infty,\Pi[s,t]} + \|f(0)\|)(t-s) \\
 &\quad + (L_g \|y\|_{\infty,\Pi[s,t]} + \|g(0)\|) \|x\|_{p,\Pi[s,t]} \\
 &\quad + KL_g \|y\|_{p,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} \\
 &\leq K \left[C_f(t-s) + L_g \|x\|_{p,\Pi[s,t]} \right] (\|y\|_{p,\Pi[s,t]} + C_0) \quad (3.11)
 \end{aligned}$$

If $C_f(t-s) + L_g \|x\|_{p,\Pi[s,t]} \leq \frac{1}{2K}$ then by taking the term containing $\|y\|_{p,\Pi[s,t]}$ to the left hand side, it follows from (3.11) that

$$\|y\|_{p,\Pi[s,t]} \leq K \left[C_f(t-s) + L_g \|x\|_{p,\Pi[s,t]} \right] (\|y\| + C_0)$$

Then

$$\begin{aligned}
& \|y\|_{p,\Pi[s,t]} + C_0 \leq \|y_s\| + C_0 + \|y\|_{p,\Pi[s,t]} \\
& \leq \left\{ 1 + K \left[C_f(t-s) + L_g \|x\|_{p,\Pi[s,t]} \right] \right\} [\|y_s\| + C_0] \\
& \leq [\|y_s\| + C_0] e^{K \left[C_f(t-s) + L_g \|x\|_{p,\Pi[s,t]} \right]}.
\end{aligned} \tag{3.12}$$

This motivates us to construct, by using the construction for the controls $\omega_{s,t}^{(1)} = C_f(t-s)$ with $\beta_1 = 1$ and $\omega_{s,t}^{(2)} = L_g^p \|x\|_{p,\Pi[s,t]}^p$ with $\beta_2 = \frac{1}{p}$, the greedy sequence of times $\tau_m = \{\tau_m(\gamma, [a, b])\}$ for the discrete time set Π on $[a, b]$ where γ is some positive constant less than $\frac{1}{2K}$. To estimate the solution norm, observe that on each $[\tau_m, \tau_{m+1}]$ such that $C_f(\tau_{m+1} - \tau_m) + L_g \|x\|_{p,\Pi[\tau_m, \tau_{m+1}]} \leq \frac{1}{2K}$, one obtains

$$\begin{aligned}
\|y_{\tau_{m+1}}\| + C_0 & \leq \|y\|_{\infty, \Pi[\tau_m, \tau_{m+1}]} + C_0 \\
& \leq \|y\|_{p, \Pi[\tau_m, \tau_{m+1}]} + C_0 \\
& \leq \exp \left\{ K \left[C_f(\tau_{m+1} - \tau_m) + L_g \|x\|_{p, \Pi[\tau_m, \tau_{m+1}]} \right] \right\} \times \\
& \quad \times (\|y_{\tau_m}\| + C_0).
\end{aligned} \tag{3.13}$$

Meanwhile, if $C_f(\tau_{m+1} - \tau_m) + L_g \|x\|_{p, \Pi[\tau_m, \tau_{m+1}]} > \gamma$ (hence τ_m, τ_{m+1} are two consecutive discrete times in Π) then it follows from (3.1) that

$$\begin{aligned}
\|y_{\tau_{m+1}}\| + C_0 &\leq C_0 + \|y_{\tau_m}\| + \|y_{\tau_m}\| \left[C_f(\tau_{m+1} - \tau_m) + L_g \|x_{\tau_m, \tau_{m+1}}\| \right] \\
&\quad + C_0 \left[C_f(\tau_{m+1} - \tau_m) + L_g \|x_{\tau_m, \tau_{m+1}}\| \right] \\
&\leq \left(1 + C_f(\tau_{m+1} - \tau_m) + L_g \|x_{\tau_m, \tau_{m+1}}\| \right) (\|y_{\tau_m}\| + C_0) \\
&\leq \exp\{C_f(\tau_{m+1} - \tau_m) + L_g \|x_{\tau_m, \tau_{m+1}}\|\} (\|y_{\tau_m}\| + C_0).
\end{aligned} \tag{3.14}$$

Combining (3.13) and (3.14) yields, for all $m = 0, \dots, N(\gamma, [a, b]) - 1$,

$$\begin{aligned}
&\|y\|_{p, \Pi[\tau_m, \tau_{m+1}]} + C_0 \\
&\leq \exp \left\{ K \left[C_f(\tau_{m+1} - \tau_m) + L_g \|x\|_{p, \Pi[\tau_m, \tau_{m+1}]} \right] \right\} (\|y_{\tau_m}\| + C_0) \\
&\leq \exp \left\{ K \left[C_f(\tau_{m+1} - \tau_m) + L_g^p \|x\|_{p, \Pi[\tau_m, \tau_{m+1}]}^p + 1 \right] \right\} (\|y_{\tau_m}\| + C_0).
\end{aligned} \tag{3.15}$$

Note that $\|x\|_{p,\Pi[s,t]}^p$ is a control, hence one can prove by induction from (3.15) that

$$\begin{aligned}
& \|y\|_{\infty,\Pi[a,b]} + C_0 \\
& \leq \exp \left\{ K \left[C_f(b-a) + L_g^p \|x\|_{p,\Pi[a,b]}^p + N(\gamma, [a, b]) \right] \right\} (\|y_a\| + C_0); \\
& \|y\|_{p,\Pi[a,b]} + C_0 \\
& \leq \|y_a\| + C_0 + N(\gamma, [a, b])^{\frac{p-1}{p}} \sum_{m=0}^{N(\gamma, [a, b])-1} \|y\|_{p,\Pi[\tau_m, \tau_{m+1}]} \\
& \leq N(\gamma, [a, b])^{\frac{2p-1}{p}} \times \tag{3.16} \\
& \quad \times \exp \left\{ K \left[C_f(b-a) + L_g^p \|x\|_{p,\Pi[a,b]}^p + N(\gamma, [a, b]) \right] \right\} (\|y_a\| + C_0),
\end{aligned}$$

which, together with (2.9), proves (3.9).

In fact, one can have better estimates with integrability of $e^{C\hat{N}(\gamma, [a, b])}$,
 where $\hat{N}$ denotes the number of times in the continuous time set, i.e.
 $\Pi = [a, b]$. (see Cass-Litterer-Lyons, 2013)

Rough DE

Next, consider the rough difference equation i.e.

$F_{s,t} = g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}$ then it follows from (3.6) and (3.10) that for generic constants $K > 1, C_0 > 0$

$$\begin{aligned}\|y\|_{p,\Pi[s,t]} &\leq (C_f\|y\|_{\infty,\Pi[s,t]} + C_0)(t-s) \\ &\quad + (L_g\|y\|_{\infty,\Pi[s,t]} + C_0)(\|x\|_{p,\Pi[s,t]} + L_g\|\mathbb{X}\|_{q,\Pi[s,t]}) \\ &\quad + K\|(y, R^y)\|_{p,\Pi[s,t]}(L_g\|x\|_{p,\Pi[s,t]} + L_g^2\|\mathbb{X}\|_{q,\Pi[s,t]}) \\ \|R^y\|_{p,\Pi[s,t]} &\leq (C_f\|y\|_{\infty,\Pi[s,t]} + C_0)(t-s) \\ &\quad + (L_g\|y\|_{\infty,\Pi[s,t]} + C_0)L_g\|\mathbb{X}\|_{q,\Pi[s,t]} \\ &\quad + K\|(y, R^y)\|_{p,\Pi[s,t]}(L_g\|x\|_{p,\Pi[s,t]} + L_g^2\|\mathbb{X}\|_{q,\Pi[s,t]}).\end{aligned}$$

it follows from the assumptions of f and g that

$$\begin{aligned}&\|(y, R^y)\|_{p,[s,t]} + C_0 \\ &\leq \|y_s\| + C_0 + K\left[C_f(t-s) + L_g\|x\|_{p,\Pi[s,t]} + L_g^2\|\mathbb{X}\|_{q,\Pi[s,t]}\right] \times \quad (3.17) \\ &\quad \times (\|(y, R^y)\|_{p,\Pi[s,t]} + C_0).\end{aligned}$$

That motivates to construct the sequence of times $\{\tau_m(\gamma, [a, b])\}$, for $\gamma \leq \frac{1}{2K}$, $\mathcal{S} = \{\omega^{(1)}, \omega^{(2)}, \omega^{(3)}\}$, $\omega_{s,t}^{(1)} = t - s$, $\beta_1 = 1$, $\omega_{s,t}^{(2)} = L_g^p \|x\|_{\rho, \Pi[s,t]}^p$, $\beta_2 = \frac{1}{p}$ and $\omega_{s,t}^{(3)} = L_g^p \|\mathbb{X}\|_{q, \Pi[s,t]}^q$, $\beta_3 = \frac{1}{q}$. The arguments are now similar to the rest of the Young case, with note of inequalities (2.10) and (3.7).

Finally, in case g is bounded, one can neglect L_g in (3.13) and the first line of the estimate (3.14) and repeat the above argument to obtain similar estimate in which ξ_1 just depends on $b - a$.

Euler scheme

Theorem: Consider the RDE

$$d\hat{y}_t = f(\hat{y}_t)dt + g(\hat{y}_t)dX_t, \hat{y}_a \in \mathbb{R}^d, t \in [a, b] \quad (3.18)$$

where X is a stochastic process with stationary increment which almost all realizations x are of local Hölder continuous with exponent in $(1/3, 1/2]$ such that x is truly rough.

If in addition to $(\mathbf{H}_f)$, $(\mathbf{H}_g)$, f and D^2g are local Lipchitz, then there exists a generic random variable $\hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b])$ such that the following estimate holds

$$\begin{aligned} & \sup_{0 \leq k \leq m} \|\hat{y}(t_k, y_a, x) - y_{t_k}\| \\ & \leq \hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b]) (|\Pi[a, b]| + \|\mathbf{x}\|_{p, [a, b], |\Pi[a, b]|})^{3-p} (b - a + \|\mathbf{x}\|_{p, [a, b]}^p). \end{aligned} \quad (3.19)$$

In particular,

$$\lim_{|\Pi[a, b]| \rightarrow 0} \sup_{0 \leq k \leq m} \|\hat{y}(t_k, \hat{y}_a, x) - y_{t_k}\| = 0. \quad (3.20)$$

Remark

It is proved in Duc-Kloeden that there exists a unique solution $\hat{y}(t, \mathbf{x}, y_a), t \in [a, b]$ to (3.18). Moreover, one can use similar estimates to prove that: there exists a generic integrable random variable $\hat{\xi}(\mathbf{x})$ such that for $0 < a < b$

$$\begin{aligned}\|\hat{y}\|_{\infty, [a, b]} &\leq (\|y_a\| + 1)\hat{\xi}(\mathbf{x}, [a, b]) \\ \|\hat{y}\|_{p, [a, b]} &\leq (\|y_a\| + 1)\hat{\xi}(\mathbf{x}, [a, b])(b - a + \|\mathbf{x}\|_{p, [a, b]} + \|\mathbb{X}\|_{q, [a, b]}) \\ \|\hat{R}\|_{q, [a, b]} &\leq (\|y_a\| + 1)\hat{\xi}(\mathbf{x}, [a, b])(b - a + \|\mathbb{X}\|_{q, [a, b]}).\end{aligned}\tag{3.21}$$

We define the numerical solution by the Euler scheme (3.1) with $y_a = \hat{y}_a$ and raise the question on the convergence of numerical solution defined in (3.1) to z . This problem is solved in Duc-Kloeden with the cut-off technique. Below we will present a direct proof using the discrete sewing lemma.

Proof

We only prove the rough case which is more difficult. Firstly, by (3.9), (3.21) one can bound $y, \hat{y}$ by a generic constant $K = K(\hat{y}_a, \mathbf{x}, [a, b])$ depends on $\hat{y}_a, \mathbf{x}, [a, b]$ and consider f, D^2g Lipchitz global with constant K for convenience.

Define $h_t = y_t - \hat{y}_t$ for all $t \in \Pi[a, b]$. Then $h_a = 0$ and

$$\begin{aligned} h_{t_k, t_{k+1}} &= [f(h_{t_k} + \hat{y}_{t_k}) - f(\hat{y}_{t_k})](t_{k+1} - t_k) + [g(h_{t_k} + \hat{y}_{t_k}) - g(\hat{y}_{t_k})]x_{t_k, t_{k+1}} \\ &\quad + [Dg(h_{t_k} + \hat{y}_{t_k})g(h_{t_k} + \hat{y}_{t_k}) - Dg(\hat{y}_{t_k})g(\hat{y}_{t_k})]\mathbb{X}_{t_k, t_{k+1}} + e_{t_k, t_{k+1}} \\ &= [f(h_{t_k} + \hat{y}_{t_k}) - f(\hat{y}_{t_k})](t_{k+1} - t_k) + F_{t_k, t_{k+1}} + e_{t_k, t_{k+1}}, \end{aligned}$$

where

- $\|e_{t_k, t_{k+1}}\| \leq \omega_{t_k, t_{k+1}}$ with control function ω of the form

$$\omega_{s, t} = K(|\Pi[a, b]| + \|\mathbf{x}\|_{p, [a, b], |\Pi[a, b]|})^{3-p} K \left[t - s + \|\mathbf{x}\|_{p, [s, t]}^p \right], \quad s < t$$

in which K is a generic constant $K = K(\hat{y}_a, \mathbf{x}, [a, b])$.

- $F_{s, t} = [g(y_s)x_{s, t} + Dg(y_s)g(y_s)\mathbb{X}_{s, t}] - [g(\hat{y}_s)x_{s, t} + Dg(\hat{y}_s)g(\hat{y}_s)\mathbb{X}_{s, t}]$

Put

$$R_{s,t}^h := h_{s,t} - [g(h_s + \hat{y}_s) - g(\hat{y}_s)] x_{s,t}, \quad (s, t) \in [a, b], s \leq t.$$

A direct computation shows that: for $s \leq u \leq t$ in $[a, b]$

$$\begin{aligned} \|(\delta F)_{s,u,t}\| &\leq \left\| \left(g(\hat{y}_u + h_u) - g(\hat{y}_u) - g(\hat{y}_s + h_s) + g(\hat{y}_s) \right. \right. \\ &\quad \left. \left. - [Dg(\hat{y}_s + h_s)g(\hat{y}_s + h_s) - Dg(\hat{y}_s)g(\hat{y}_s)]x_{s,u} \right) \otimes x_{u,t} \right\| \\ &\quad + \|Dg(\hat{y}_u + h_u)g(\hat{y}_u + h_u) - Dg(\hat{y}_u)g(\hat{y}_u) \\ &\quad - Dg(\hat{y}_s + h_s)g(\hat{y}_s + h_s) + Dg(\hat{y}_s)g(\hat{y}_s)\| \|\mathbb{X}_{u,t}\|. \end{aligned}$$

Due to the assumption $(\mathbf{H}_g)$, it is easy to show that $Dg(y)g(y)$ is Lipschitz continuous with the Lipschitz constant KL_g for a generic constant $K > 0$, thus the second term in the above inequality can be estimated as

$$\begin{aligned} &\|Dg(\hat{y}_u + h_u)g(\hat{y}_u + h_u) - Dg(\hat{y}_u)g(\hat{y}_u) - Dg(\hat{y}_s + h_s)g(\hat{y}_s + h_s) \\ &\quad + Dg(\hat{y}_s)g(\hat{y}_s)\| \|\mathbb{X}_{u,t}\| \leq KL_g (\|h_{s,u}\| + \|h\|_{\infty,[s,u]} \|\hat{y}_{s,u}\|) \|\mathbb{X}_{u,t}\| \\ &\leq KL_g \left[\|h\|_{p,\Pi[s,t]} + \|h\|_{\infty,\Pi[s,t]} (\|\hat{y}\|_{p,\Pi[s,t]} + \|y\|_{p,\Pi[s,t]}) \right] \|\mathbb{X}\|_{q,\Pi[s,t]}. \end{aligned}$$

Meanwhile, the first term has the form $\|M \otimes x_{u,t}\|$ where

$$\begin{aligned}
\|M \otimes x_{u,t}\| &= \left\| \int_0^1 \left(Dg(\hat{y}_s + h_s + \eta(\hat{y}_{s,u} + h_{s,u}))(\hat{y}_{s,u} + h_{s,u}) \right. \right. \\
&\quad \left. \left. - Dg(\hat{y}_s + \eta\hat{y}_{s,u})\hat{y}_{s,u} \right) d\eta \right. \\
&\quad \left. - [Dg(\hat{y}_s + h_s)g(\hat{y}_s + h_s) - Dg(\hat{y}_s)g(\hat{y}_s)]x_{s,u} \right\| \|x_{u,t}\| \\
&\leq K \left(\|R^h\|_{q,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} + \|h\|_{p,\Pi[s,t]} \|\mathbb{X}\|_{q,\Pi[s,t]} \right) \\
&\quad + K \|h\|_{\infty,\Pi[s,t]} \left(\|x\|_{p,\Pi[s,t]} + \|\mathbb{X}\|_{q,\Pi[s,t]} \right).
\end{aligned}$$

As a result,

$$\begin{aligned}
\|h_{s,t}\| &\leq K \|h\|_{\infty,\Pi[s,t]}(t-s) + K \|h\|_{\infty,\Pi[s,t]}(\|x_{s,t}\| + \|\mathbb{X}_{s,t}\|) + \omega_{s,t} \\
&\quad + KL_g \left(\|R^h\|_{q,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} + \|h\|_{p,\Pi[s,t]} \|\mathbb{X}\|_{q,\Pi[s,t]} \right); \\
\|R_{s,t}^h\| &\leq C_f \|h\|_{\infty,\Pi[s,t]}(t-s) + KL_g \|h\|_{\infty,\Pi[s,t]} \|\mathbb{X}_{s,t}\| + \omega_{s,t} \\
&\quad + K \left(\|R^h\|_{q,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} + \|h\|_{p,\Pi[s,t]} \|\mathbb{X}\|_{q,\Pi[s,t]} \right).
\end{aligned}$$

One can now apply the same arguments above to show that there exists a random variable $\hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b])$ such that

$$\begin{aligned} \|h\|_{\infty, [a, b]} &\leq \|(h, R^h)\|_{p, [a, b]} \leq (\|h_a\| + \omega_{a, b}) \hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b]) \\ &\leq K(|\Pi[a, b]| + \|\mathbf{x}\|_{p, [a, b], |\Pi[a, b]|})^{3-p} \end{aligned}$$

which proves (3.19) for $\hat{\xi}$ generic. Finally, because $p \in [2, 3)$ and $\mathbf{x} \in C^{\frac{1}{p}}$ is of $\frac{1}{p}$ -Hölder rough path, it is easy to check that $\|\mathbf{x}\|_{p, [a, b], |\Pi[a, b]|} \rightarrow 0$ as $|\Pi[a, b]| \rightarrow 0$. This proves (3.20).

Probablistic setting

$p \in (0, 3)$ and a regular discrete time set $\Pi^\Delta = \{k\Delta : k \in \mathbb{Z}\}$, denote by $\mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ the space of all paths $\mathbf{g} : \Pi^\Delta \rightarrow T_1^2(\mathbb{R}^m)$. Then $\mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ is equipped with the compact open topology given by the p -variation norm (2.2), i.e the topology generated by the metric

$$d_p^\Delta(\mathbf{g}, \mathbf{h}) := \sum_{k \geq 1} \frac{1}{2^k} (\|\mathbf{g} - \mathbf{h}\|_{p, \Pi[-k\Delta, k\Delta]} \wedge 1).$$

As a result, it is separable and thus a Polish space.

Let us consider a stochastic process $\bar{\mathbf{X}}$ defined on a probability space $(\bar{\Omega}, \bar{\mathcal{F}}, \bar{\mathbb{P}})$ with realizations in $(\mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m)), \mathcal{F})$. Assume further that $\bar{\mathbf{X}}$ has stationary increments. Assign $\Omega := \mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ and equip it with the Borel σ -algebra $\mathcal{F}$ and let $\mathbb{P}$ be the law of $\bar{\mathbf{X}}$.

Denote by θ the *Wiener-type shift*

$$(\theta_t \omega)_\cdot = \omega_t^{-1} \bullet \omega_{t+\cdot}, \quad \forall t \in \Pi^\Delta, \omega \in \mathcal{C}_0^p(\Pi^\Delta, \mathcal{T}_1^2(\mathbb{R}^m)), \quad (4.1)$$

and define the so-called *diagonal process*

$\mathbf{X} : \Pi^\Delta \times \Omega \rightarrow \mathcal{T}_1^2(\mathbb{R}^m)$, $\mathbf{X}_t(\omega) = \omega_t$ for all $t \in \Pi^\Delta, \omega \in \Omega$. Due to the stationarity of $\bar{\mathbf{X}}$, it can be proved that θ is invariant under $\mathbb{P}$, then forming a discrete (and thus measurable) dynamical system on $(\Omega, \mathcal{F}, \mathbb{P})$, Theorem 5-(BRSch17). Moreover, $\mathbf{X}$ forms an *p-rough path cocycle*, namely, $\mathbf{X}_\cdot(\omega) \in \mathcal{C}_0^p(\Pi^\Delta, \mathcal{T}_1^2(\mathbb{R}^m))$ for every $\omega \in \Omega$, which satisfies the *cocycle relation*:

$$\mathbf{X}_{t+s}(\omega) = \mathbf{X}_s(\omega) \bullet \mathbf{X}_t(\theta_s \omega), \forall \omega \in \Omega, t, s \in \Pi^\Delta,$$

in the sense that $\mathbf{X}_{s,s+t} = \mathbf{X}_t(\theta_s \omega)$ with the increment notation $\mathbf{X}_{s,s+t} := \mathbf{X}_s^{-1} \bullet \mathbf{X}_{s+t}$. In particular, the Wiener shift (4.1) implies that

$$\|\mathbf{x}(\theta_h \omega)\|_{p, \Pi[s, t]} = \|\mathbf{x}(\omega)\|_{p, \Pi[s+h, t+h]}, \quad \forall s, t, h \in \Pi^\Delta. \quad (4.2)$$

Discrete RDS

Define the discrete mapping $H^\Delta(\mathbf{x})y = y + f(y)\Delta + F_{0,\Delta}(y, \mathbf{x})$. Similar to Duc-Kloeden, we can easily prove that the discrete Euler scheme (3.1) generates a discrete random dynamical system $\varphi^\Delta : \Pi^\Delta \times \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ over $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ such that

$$\begin{aligned}\varphi^\Delta(0, \omega)y_0 &= y_0, \\ \varphi^\Delta(k\Delta, \omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega) \circ \dots \circ H^\Delta(\omega)y_0, \quad \forall k \geq 1.\end{aligned}\tag{4.3}$$

Throughout this paper, we assume that θ is an ergodic dynamical system.

Ergodicity of θ

- Garrido-Atienza, B. Schmalfuss (2011): the Wiener shift $\eta_t x = x_{t+} - x_t$ is ergodic on $\Omega' = C^{0,\beta}(\mathbb{R}, \mathbb{R}^m)$ w.r.t. $\mathbb{P}_H$.
- Fiz-Hairer (2014, Theorem 10.4): there exists a full measure subset $\Omega_1 \subset \Omega'$ such that $\omega = (1, x, \mathbb{X}) \in \Omega$ for any $x \in \Omega_1$. Moreover, by Friz-Victoir (2010, Theorem 15.42, 45), one can choose this full measure subset Ω_1 such that it satisfies the piecewise linear approximations (mollifier approximation).
- Then consider the natural lift $\mathcal{S}$ on smooth paths

$$\mathcal{S}(x)_{s,t} = (1, x_{s,t}, \int_s^t x_{s,r} dx_r), \quad (4.4)$$

which can be extended to Ω_1 such that $\mathcal{S} : \Omega_1 \rightarrow \Omega$. By Bailleul, Riedel, Scheutzow (2017), there exists a metric dynamical system $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}}, \theta)$ such that $\hat{\Omega} \subset \Omega$ and $\hat{\mathbb{P}} = \mathbb{P}^H \circ \mathcal{S}^{-1}$. Furthermore,

$$\theta_t \mathcal{S}(x) = \lim_{n \rightarrow \infty} \hat{\theta} \mathcal{S}(x^{(n)}) = \lim_{n \rightarrow \infty} \mathcal{S}(\eta_t x^{(n)}) = \mathcal{S}(\eta_t x). \quad (4.5)$$

- Since η is ergodic, by Garrido-Atienza, B. Schmalfuss (2011, Lemma 3), θ is also ergodic.
- From now on, we assume additionally that $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ is ergodic.

Pullback attractors

Roughly speaking, an invariant random compact set $\mathcal{A} \in \mathcal{D}$ is called a *pullback attractor* in $\mathcal{D}$, if $\mathcal{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the pullback sense, i.e.

$$\lim_{t \rightarrow \infty} d_H(\varphi(t, \theta_{-t}\omega) \hat{D}(\theta_{-t}\omega) | \mathcal{A}(\omega)) = 0, \quad (4.6)$$

where $d_H(\cdot | \cdot)$ is the Hausdorff semi-distance, i.e.

$d_H(D | A) := \sup_{d \in D} \inf_{a \in A} \|d - a\|$. A random set $\mathcal{B} \in \mathcal{D}$ is called *pullback absorbing* in the universe $\mathcal{D}$ if $\mathcal{B}$ absorbs all closed random sets in $\mathcal{D}$, i.e. for any closed random set $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega) \hat{D}(\theta_{-t}\omega) \subset \mathcal{B}(\omega), \text{ for all } t \geq t_0. \quad (4.7)$$

$$\mathcal{A}(\omega) = \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} \varphi(s, \theta_{-s}\omega) \mathcal{B}(\theta_{-s}\omega)}. \quad (4.8)$$

Lemma

(i) Let $a : \Omega \rightarrow [0, \infty)$ be a random variable, $\log(1 + a(\cdot)) \in L^1$ and $\hat{a} := E \log(1 + a(\cdot)) = \int_{\Omega} \log(1 + a(\cdot)) d\mathbb{P}$. Let $\lambda > \hat{a}$ be an arbitrary fixed positive number. Put

$$b(x) := \sum_{k=1}^{\infty} e^{-\lambda k} \prod_{i=0}^{k-1} (1 + a(\theta_{-i}x)).$$

Then $b(\cdot)$ is a nonnegative almost everywhere finite and tempered random variable.

(ii) Let $c : \Omega \rightarrow [0, \infty)$ be a tempered random variable, and $\delta > 0$ be an arbitrary fixed positive number. Put

$$d(x) := \sum_{k=1}^{\infty} e^{-\delta k} c(\theta_{-k}x).$$

Then $d(\cdot)$ is a nonnegative almost everywhere finite and tempered random variable.

Existence theorem

Now we assume further that f is dissipative: there exist $c, d > 0$ so that for all $y \in \mathbb{R}^d$,

$$\langle y, f(y) \rangle \leq c - d\|y\|^2. \quad (4.9)$$

In addition, assume the p -variation norm of $\mathbf{x}$ is of finite moment for all order $k \in \mathbb{N}$, i.e.

$$\mathbb{E} \|\mathbf{x}(\cdot)\|_{p, \Pi[a, b]}^k < \infty, \quad \forall k \in \mathbb{N}, \quad \forall 0 < a < b, \quad a, b \in \Pi. \quad (4.10)$$

Assume $(\mathbf{H}_f)$, $(\mathbf{H}_g)$ and the dissipativity condition (4.9) and condition (4.10). Consider system (3.1) with the regular time set Π^Δ , where $\Delta \in (0, 1)$ satisfied the inequality

$$0 < \Delta < 1 \wedge \frac{d}{2C_f^2} \wedge \frac{1}{2d}. \quad (4.11)$$

Then for L_g small enough, the generated discrete random dynamical system φ^Δ of (3.1) admits a random pullback attractor $\mathcal{A}^\Delta$.

Proof

Given $\Delta \in (0, 1)$ small enough, one introduces $n \in \mathbb{N}$ such that $T := n\Delta \leq 1 < (n+1)\Delta$. First, one applies the Lyapunov function $\|y\|^2$ into (3.1) and uses Cauchy-Schwarz inequality to obtain for $t_k = k\Delta$, $k = 0, \dots, (n-1)$

$$\begin{aligned} \|y_{t_{k+1}}\|^2 &= \|y_{t_k}\|^2 + 2\langle y_{t_k}, f(y_{t_k}) \rangle \Delta + \|f(y_{t_k})\|^2 \Delta^2 + 2\langle y_{t_k}, F_{t_k, t_{k+1}} \rangle \\ &\quad + 2\langle f(y_{t_k}), F_{t_k, t_{k+1}} \rangle \Delta + \|F_{t_k, t_{k+1}}\|^2 \\ &\leq \|y_{t_k}\|^2 (1 - 2d\Delta + 2C_f^2 \Delta^2) + K(\Delta + \Delta^2) \\ &\quad + 2\|f(y)\|_{\infty, \Pi[0, T]} \|F\|_{\infty, \Pi[0, T]} \Delta \\ &\quad + 2\langle y_{t_k}, F_{t_k, t_{k+1}} \rangle + \|F_{t_k, t_{k+1}}\|^2. \end{aligned} \tag{4.12}$$

It follows from (4.11) that

$$0 < 1 - 2d\Delta + 2C_f^2\Delta^2 < 1 - d\Delta < e^{-d\Delta}.$$

Then (4.12) leads to, for a generic constant $K > 1$, the following estimate

$$\begin{aligned} \|y_{t_{k+1}}\|^2 &\leq e^{-d\Delta} \|y_{t_k}\|^2 + [2\langle y_{t_k}, F_{t_k, t_{k+1}} \rangle + \|F_{t_k, t_{k+1}}\|^2] \\ &\quad + K\Delta(1 + \|f(y)\|_{\infty, \Pi[0, T]} \|F\|_{\infty, \Pi[0, T]}). \end{aligned}$$

Assign $G_{s,t} := 2\langle y_s, F_{s,t} \rangle + \|F_{s,t}\|^2$. By induction one can prove that

$$\begin{aligned} \|y_{t_n}\|^2 &\leq e^{-n\Delta d} \|y_0\|^2 + K\Delta (1 + \|f(y)\|_{\infty, \Pi[0, T]} \|F\|_{\infty, \Pi[0, T]}) \times \\ &\quad \times \left(\sum_{k=0}^{n-1} e^{-d(n-k)\Delta} \right) + \sum_{k=0}^{n-1} e^{-d(t_n - t_k)} G_{t_k, t_{k+1}} \\ &\leq e^{-dT} \|y_0\|^2 + K (1 + \|f(y)\|_{\infty, \Pi[0, T]} \|F\|_{\infty, \Pi[0, T]}) \\ &\quad + \sum_{k=0}^{n-1} e^{-d(t_n - t_k)} G_{t_k, t_{k+1}}. \end{aligned} \tag{4.13}$$

Define $\tilde{G}_{s,t} := e^{-d(t_n-s)} G_{s,t}$. Observe that

$$\begin{aligned} \|(\delta \tilde{G})_{s,u,t}\| &\leq (e^{-d(t_n-s)} - e^{-d(t_n-u)}) \|G_{u,t}\| + \|e^{-d(t_n-s)}\| \|(\delta G)_{s,u,t}\| \\ &\leq (t-s) \|F\|_{\infty, \Pi[s,t]} \left(2\|y\|_{\infty, \Pi[0,T]} + \|F\|_{\infty, \Pi[0,T]} \right) \\ &\quad + \|(\delta G)_{s,u,t}\|, \quad 0 \leq s \leq u \leq t \leq T \end{aligned}$$

where the first term has the form of a control w.r.t. a suitable $\lambda > 1$.
The second term is estimated as follows

$$\begin{aligned} &\|(\delta G)_{s,u,t}\| \\ &\leq \left\| 2\langle y_s, F_{s,t} - F_{s,u} \rangle - 2\langle y_u, F_{u,t} \rangle + 2\langle F_{s,u}, F_{u,t} \rangle \right. \\ &\quad \left. + 2\langle F_{s,u} + F_{u,t}, (\delta F)_{s,u,t} \rangle + [(\delta F)_{s,u,t}]^2 \right\| \\ &\leq \left\| 2\langle y_s, \delta F_{s,u,t} \rangle - \langle y_{s,u} - F_{s,u}, F_{u,t} \rangle + \langle F_{s,t} + F_{s,u} + F_{u,t}, (\delta F)_{s,u,t} \rangle \right\| \\ &\leq \omega_{s,t}^{\lambda_F}(\delta F) \left(2\|y\|_{\infty, \Pi[0,T]} + 3\|F\|_{\infty, \Pi[0,T]} \right) \\ &\quad + 2 \left(\|f(y)\|_{\infty, \Pi[0,T]}(t-s) + K\omega_{s,t}^{\lambda_F}(\delta F) \right) \|F\|_{\infty, \Pi[0,T]} \end{aligned}$$

As a result, $\delta \tilde{G}_{s,u,t}$ is bounded by a control up to a power $\lambda_{\tilde{G}}$ with

$$\begin{aligned} \omega_{s,t}^{\lambda_{\tilde{G}}}(\delta \tilde{G}) &\leq K(t-s)\|F\|_{\infty,[s,t]}\left(\|y\|_{\infty,\Pi[0,T]} + \|F\|_{\infty,\Pi[0,T]} + 1\right) \\ &\quad + K\omega_{s,t}^{\lambda_F}(\delta F)\left(\|y\|_{\infty,\Pi[0,T]} + \|F\|_{\infty,\Pi[0,T]}\right). \end{aligned} \quad (4.14)$$

Therefore, by applying the discrete sewing lemma to (4.13), one obtains, for a generic constant K

$$\begin{aligned} \|y_T\|^2 &\leq e^{-dT}\|y_0\|^2 + K\left(1 + \|f(y)\|_{\infty,\Pi[0,T]}\|F\|_{\infty,\Pi[0,T]}\right) \\ &\quad + e^{-dT}G_{0,T} + K\omega_{0,T}^{\lambda_{\tilde{G}}}(\delta \tilde{G}) \\ &\leq e^{-dT}\|y_0\|^2 \\ &\quad + K\left[1 + (\|y\|_{\infty,\Pi[0,T]} + C_0)\|F\|_{\infty,\Pi[0,T]} + \|F\|_{\infty,\Pi[0,T]}^2\right] \\ &\quad + K\omega_{s,t}^{\lambda_F}(\delta F)\left(\|y\|_{\infty,\Pi[0,T]} + \|F\|_{\infty,\Pi[0,T]}\right). \end{aligned} \quad (4.15)$$

One can prove that

$$\|F\|_{\infty, \Pi[s, t]} \leq (L_g \|y\|_{\infty, \Pi[s, t]} + C_0) \|\mathbf{x}\|_{p, \Pi[s, t]}$$

one concludes that there exists generic constants $C_0 > 0$, a generic polynomial ξ_1 , and a random variable ξ_2 of the form $\xi_2(z) = z^m$, $z \in \mathbb{R}^+$ for some (fractional order) $m \geq p$ such that (4.14) has the form

$$\begin{aligned} \|y_T\|^2 &\leq e^{-dT} \left(1 + L_g \xi_1(\|\mathbf{x}\|_{p, \Pi[0, T]}) e^{\xi_2(L_g \|\mathbf{x}\|_{p, \Pi[0, T]})} \right) \|y_0\|^2 \\ &\quad + C_0 \xi_1(\|\mathbf{x}\|_{p, \Pi[0, T]}) e^{\xi_2(L_g \|\mathbf{x}\|_{p, \Pi[0, T]})} \end{aligned} \quad (4.16)$$

One now applies the inequality $\log(1 + \mu e^\nu) \leq \mu + \nu$, $\forall \mu, \nu \in \mathbb{R}, \nu \geq 0$, to obtain

$$\begin{aligned} &\log \left(1 + L_g \xi_1(\|\mathbf{x}\|_{p, \Pi[0, T]}) e^{\xi_2(L_g \|\mathbf{x}\|_{p, \Pi[0, T]})} \right) \\ &\leq L_g \xi_1(\|\mathbf{x}\|_{p, \Pi[0, T]}) + \xi_2(L_g \|\mathbf{x}\|_{p, \Pi[0, T]}) \in L^1. \end{aligned} \quad (4.17)$$

Hence, one can choose L_g small enough such that

$$L_g \mathbb{E} \xi_1(\|\mathbf{x}(\cdot)\|_{p, \Pi[0, T]}) + \mathbb{E} \xi_2(L_g \|\mathbf{x}(\cdot)\|_{p, \Pi[0, T]}) < dT. \quad (4.18)$$

Similarly to Cong-Duc-Hong (2020), one can use (4.16), (4.17) to prove that there exists an absorbing set of the form $B^\Delta(\omega) = \bar{B}(0, R^\Delta(\mathbf{x}(\omega)))$, where

$$\begin{aligned} R^\Delta(\mathbf{x}) &= 1 + C_0 \sum_{k \geq 1} e^{-dT_k} \Lambda(\mathbf{x}, \Pi[-kT, -(k-1)T]) \\ &\quad \times \prod_{i=1}^k (1 + L_g \Lambda(\mathbf{x}, \Pi[-iT, -(i-1)T])) \end{aligned} \quad (4.19)$$

with $T = n\Delta$ and $\Lambda(\mathbf{x}, \Pi[a, b]) = \xi_1(\|\mathbf{x}\|_{p, \Pi[a, b]}) e^{\xi_2(L_g \|\mathbf{x}\|_{p, \Pi[a, b]})}$. Hence, there exists a random pullback attractor $\mathcal{A}^\Delta(\omega)$ which is contained in $B^\Delta(\omega)$.

Remark

Next we recall that in Duc-Kloeden, it is proved that if f satisfies dissipative condition, (3.18) generate a RDS φ which possesses a pullback random attractor $\mathcal{A}$. At the same time (3.18) generate a discrete RDS which possesses a random pullback attractor $\mathcal{A}^\Delta$ when $\Delta \leq \Delta_0$ small enough. In the following we prove that the discrete attractor converges to the continuous attractor as Δ approaches 0. Note that the result here is obtained without the condition on boundedness of f as proved in Duc-Kloeden.

To satisfy condition (4.10), we assume a stronger condition that

$$\mathbb{E} \|\mathbf{x}\|_{p,[a,b]}^k < \infty, \quad \forall k \in \mathbb{N}, \quad \forall 0 < a < b. \quad (4.20)$$

Proposition: There exists $L > 0$ such that if $L_g < L$, for all Δ small enough the attractor $\mathcal{A}^\Delta$ of (3.1) is uniformly bounded in Δ .

Proof of Proposition

For T and $R^\Delta(\mathbf{x})$ where Δ small enough so that $T > 1/2$, observe that for each k , $n_k = \lfloor kT \rfloor$ satisfies $n_k \leq kT < (k+1)T < n_k + 2$ and $n_k \leq n_{k+1} \leq n_k + 1$, $n_k < n_{k+2}$. We then have some generic constant M (note that $n_k \leq k$)

$$\begin{aligned}
 & R^\Delta(\mathbf{x}) \\
 & \leq 1 + M \sum_{k=0}^{\infty} e^{-dk} \Lambda(\mathbf{x}, [-k, -k+2]) \prod_{j=0}^k \left[1 + L_g \Lambda(\mathbf{x}, [-j-1, -j+1]) \right]^2 \\
 & \leq 1 + M \sum_{k=1}^{\infty} e^{-dk} \Lambda(\theta_{-k}\mathbf{x}, [-1, 2]) \prod_{j=1}^k \left[1 + L_g \Lambda(\theta_{-j}\mathbf{x}, [-1, 2]) \right]^2 =: R(\mathbf{x}).
 \end{aligned}
 \tag{4.21}$$

Choose $L_g \ll 1$ independent of Δ st (4.21) converges almost all.
 $\Rightarrow \mathcal{A}^\Delta \subset \bar{B}(0, R(\mathbf{x}))$ where the radius $R(\mathbf{x})$ is independent of Δ .

Semi-continuity of attractors

Theorem: There exists $L > 0$ such that if $L_g < L$, for all Δ small enough the attractor $\mathcal{A}^\Delta$ of (3.1) is uniformly bounded in Δ .

As a result, for L_g small enough, the numeric attractor $\mathcal{A}^\Delta$ converges to the attractor $\mathcal{A}$ in the Hausdorf semi-distance, i.e. $d_H(\mathcal{A}^\Delta|\mathcal{A}) \rightarrow 0$ as $\Delta \rightarrow 0$, a.s.

Proof

First, for L_g small enough so that $R(\mathbf{x})$ is finite, we have $R^\Delta(\mathbf{x}) \leq R(\mathbf{x})$ for all $\Delta < \Delta_0$ small.

Now, we proceed a contradiction arguments. Namely, assume the assertion is false for a particular ω corresponding to a rough path $\mathbf{x}$, then there exists an $\varepsilon_0 > 0$ and a sequence $\Delta_j \downarrow 0^+$ such that $d_H(\mathcal{A}^{\Delta_j}(\omega)|\mathcal{A}(\omega)) > \varepsilon_0$ for all $j \in \mathbb{N}$. Since these attractors are compact sets, there exists $a_j(\omega) \in \mathcal{A}^{\Delta_j}(\omega)$ such that $d_H(a_j(\omega)|\mathcal{A}(\omega)) > \varepsilon_0$. Due to the invariance, there exists for each $m_j \in \mathbb{N}$ a point $b_j(\theta_{-m_j\Delta_j}\omega) \in \mathcal{A}^{\Delta_j}(\theta_{-m_j\Delta_j}\omega) \subset \mathcal{B}^{\Delta_j}(\theta_{-m_j\Delta_j}\omega)$ such that $\varphi^{\Delta_j}(m_j\Delta_j, \theta_{-m_j\Delta_j}\omega)b_j(\theta_{-m_j\Delta_j}\omega) = y_{m_j\Delta_j}^{\Delta_j}(\theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega)) = a_j(\omega)$. Respectively one considers the continuous flow with respect to the rough differential equation

$$\varphi(m_j\Delta_j, \theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega)) = \hat{y}_{m_j\Delta_j}(\theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega))$$

and applies the triangle inequality to obtain

$$\begin{aligned}
\varepsilon_0 &< d_H(a_j(\omega)|\mathcal{A}(\omega)) \\
&\leq \|\hat{y}_{m_j\Delta_j}(\theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega)) - y_{m_j\Delta_j}^{\Delta_j}(\theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega))\| \\
&\quad + d_H(\varphi(m_j\Delta_j, \theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega))|\mathcal{A}(\omega)). \tag{4.22}
\end{aligned}$$

On the other hand, since $\mathcal{A}$ attracts $\bar{B}(0, R(\omega))$, i.e. there exists a fixed $T^* := T(\varepsilon_0, \omega)$ such that for any $m_j\Delta_j \in [T^*, T^* + 1]$

$$d_H(\varphi(m_j\Delta_j, \theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega))|\mathcal{A}(\omega)) \leq \frac{\varepsilon_0}{2}. \tag{4.23}$$

Meanwhile, due to the estimate (3.19), the first difference in the right hand side of (4.22), for a fixed interval $[0, T^* + 1]$, is uniform in $\mathcal{B}^{\Delta_j}(\theta_{-m_j\Delta_j}\omega)$. Hence one obtains for Δ_j small enough

$$\begin{aligned}
&\|\hat{y}_{m_j\Delta_j}(\theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega)) - y_{m_j\Delta_j}^{\Delta_j}(\theta_{-m_j\Delta_j}\omega, b_j(\theta_{-m_j\Delta_j}\omega))\| \\
&\leq K(R^{\Delta_j}(\theta_{-m_j\Delta_j}\omega)) \left[\Delta_j + \|\omega\|_{\rho, [-T^*-1, 1], \Delta_j} \right]^{3-p} \\
&\quad \times \left[T^* + 1 + \|\omega\|_{\rho, [-T^*-1, 1]}^p \right]. \tag{4.24}
\end{aligned}$$

Write $m_j \Delta_j = T^* + s_j$ with $s_j \in [0, 1]$, one has

$$\begin{aligned}
 R^{\Delta_j}(\theta_{-m_j \Delta_j} \omega) &\leq 1 + M \sum_{k=0}^{\infty} e^{-dk} \Lambda(\theta_{-k} \theta_{-T^*} x, [-1, 2]) \times \\
 &\quad \times \prod_{j=0}^k \left[1 + L_g \Lambda(\theta_{-j} \theta_{-T^*} \omega, [-1, 2]) \right]^2 \\
 &\leq R(\theta_{-T^*}(\omega)).
 \end{aligned} \tag{4.25}$$

Hence, the right hand side of (4.24) tends to 0 as $j \rightarrow \infty$, i.e.

$$\|\hat{y}_{m_j \Delta_j}(\theta_{-m_j \Delta_j} \omega, b_j(\theta_{-m_j \Delta_j} \omega)) - y_{m_j \Delta_j}^{\Delta_j}(\omega, b_j(\theta_{-m_j \Delta_j} \omega))\| \leq \frac{\varepsilon_0}{2} \tag{4.26}$$

for j large enough. Since (4.23) and (4.26) contradict to (4.22), this completes the proof.

Proof for bounded g

Here we would like to discuss about the case g bounded. As mentioned earlier, the existence of discrete random attractor for system (3.1) is proved in Duc-Kloeden for any L_g and Δ small enough. Under a stronger condition that

$$\mathbb{E} \|\mathbf{x}\|_{1/p\text{-Hol},[a,b]}^k < \infty, \quad \forall k \in \mathbb{N}, \quad \forall 0 < a < b, \quad (4.27)$$

we can prove for arbitrary L_g . Indeed, we construct the sequence $\tau_m(\delta, [0, T])$ on $[0, T]$ and obtain the estimate

$$\sum_{m=0}^{N(\delta, [0, T])-1} \|y, R^y\|_{p, \Pi[\tau_m, \tau_{m+1}]} \leq K \left[T \|y\|_{\infty, \Pi[0, T]} + \xi(\|\mathbf{x}\|_{p, \Pi[0, T]}) \right]$$

for a sufficiently small δ , a generic polynomial ξ and a generic constant K .

We modify (4.13) as

$$\begin{aligned}
\|y_{t_n}\|^2 &\leq e^{-d}\|y_0\|^2 + K \left(1 + \|f(y)\|_{\infty, \Pi[0, T]}\|F\|_{\infty, \Pi[0, T]}\right) \\
&\quad + \sum_{k=0}^{n-1} e^{-d(t_n - t_k)} G_{t_k, t_{k+1}} \\
&\leq e^{-dT}\|y_0\|^2 + K \left(1 + \|f(y)\|_{\infty, \Pi[0, T]}\|F\|_{\infty, \Pi[0, T]}\right) \\
&\quad + \sum_{i=0}^{N(\delta, [0, T])-1} \left[G_{\tau_i, \tau_{i+1}} + K \omega_{\tau_i, \tau_{i+1}}^{\lambda_{\tilde{G}}}(\delta \tilde{G}) \right].
\end{aligned} \tag{4.28}$$

Since g is bounded, we use the estimate $\|F\|_{\infty, \Pi[s, t]} \leq L_g \|\mathbf{x}\|_{p, \Pi[s, t]}$ and choose $\delta = (1 \wedge d(1 - \Delta_0))/2K$. Then a direct calculation leads to

$$\|y_T\|^2 \leq e^{-dT/2} (1 + L_g \Delta^{1/p} \|\mathbf{x}\|_{1/p - \text{Hol}, \Pi[0, T]}) \|y_0\|^2 + K e^{\xi(\|\mathbf{x}\|_{1/p - \text{Hol}, \Pi[0, T]})},$$

which is similar to (4.16). This proves the theorem for arbitrary L_g .

References

1. Christian Bayer, Peter Friz, Nikolas Tapia. Stability of deep neural network via discrete rough paths. SIAM J. Math. Data Science, 2023.
2. I. Bailleul, S. Riedel, M. Scheutzow. Random dynamical systems, rough paths and rough flows. JDE 2017.
3. Th. Cass, C. Litterer, T. Lyons. Integrability and tail estimates for Gaussian rough differential equations. Ann. Prob. 2013.
4. Nguyen Dinh Cong, LHD, Phan Thanh Hong. Numerical attractors via discrete rough paths. submitted, 2022.
5. LHD, Peter Kloeden. Numerical attractors for rough differential equations. MIS preprint 2021.
6. LHD. Random attractors for dissipative systems with rough noises. DCDS-A, 2022.
7. Nguyen Dinh Cong, LHD, Phan Thanh Hong. Pullback attractors for stochastic Young differential delay equations. JDDE, 2022.
8. M.J. Garrido-Atienza, B. Schmalfuss. Ergodicity of the infinite dimensional fractional brownian motion. JDDE, 2011.